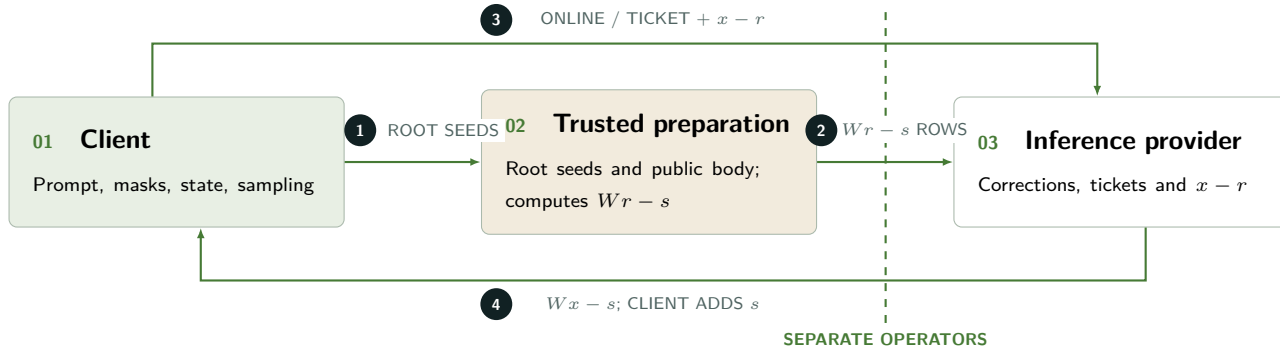


Private inference for an open compute world

Private inference can let organizations use remote AI compute without exposing prompts, context, or generated output to the infrastructure provider. PLLM keeps sensitive data on the client and sends protected requests to remote services. This design could support a broader market of independent compute without requiring users to trust one provider with all of their data.



What PLLM changes

Ordinary hosted inference bundles computation with access to prompts and generated text. PLLM instead keeps tokenization, nonlinear operations, attention state, sampling, and decoding in the client environment. Public transformer-body matrices run remotely as masked integer projections. Compute can therefore be selected independently from willingness to disclose client language. This is a protocol boundary, not a URL setting.^[1]

Prepared inventory

Before chat, the client commits an inventory to one model, body, stage set, quantization profile, and attempt budget. It sends one domain-separated root seed per stage to preparation. Preparation expands one-time input masks r , output masks s , and tickets, computes $Wr - s$, then pushes each batch to inference. Inference acknowledges accepted stage batches and seals the inventory READY. Online, it atomically consumes a ticket and returns

$$W(x - r) + (Wr - s) = Wx - s.$$

The client adds s . Prefill rows share a compact matrix envelope; decode retains one ticket per stage on a persistent connection. Unused reserved rows burn on cancellation or failure. Inventory is memory-only and disappears on restart or idle expiry.^[1]

Boundary, not invisibility

Protected from inference: plaintext prompts and outputs, token identities, root seeds, masks, activation scales, local attention and recurrent state, and sampling choices. Preparation sees seeds and bound stage metadata, but not online masked activations. Inference sees corrections and masked activations, but not seeds.

Still exposed: both services know the public model, stage names, tensor shapes, quantization profile, scheduling, timing, traffic volume, and approximate sequence lengths. Preparation must follow the protocol, erase expanded masks, and not collude with inference. Two services under one untrusted operator do not meet this assumption; authentication and TLS protect channels but do not prove independence, erasure, or correct model execution.^[2]

SHAKE-256 expands each fresh root seed into pseudorandom masks. Under that computational assumption, each masked activation hides its input from inference when viewed alone. The claim assumes protocol-conforming roles, mask non-retention, and non-collusion; Python releases masks after use but verified physical zeroization is not established. Deployment review must therefore name operators and log-access paths, not merely count endpoints.

A world of compute, without exposing client data.



Unlock untrusted capacity

Hosted inference usually requires the buyer to trust the compute owner with plaintext. Masked execution separates those choices. Regional clouds, renewable projects, sovereign infrastructure, colocation operators, and independent machines could compete without becoming custodians of client language. Participation still requires compatible execution, availability, credentials, and enforceable market rules—but not permission to read the workload.

Follow available energy

Solar generation, grid congestion, and idle capacity vary by place and hour. A scheduler could move masked public-model work toward abundant renewable generation and replenish of-line inventory where latency matters less. Privacy makes broader routing plausible; it does not make it free. Dependent client exchanges, correction placement, distance, and operator separation remain physical constraints. PLLM does not yet measure an energy saving.

Price the complete lifecycle

Client	Local model work, state, memory and traffic
Preparation	Body storage, matrix work, correction egress and erasure
Inference	Masked compute, inventory, traffic and tickets
Market	Discovery, settlement, verification and operator separation

Broader supply can create price pressure and wider access to private AI. This is a market mechanism, not a measured savings result.

No plaintext training feed

On the current path, Inference receives one-time tickets and $x-r$; Preparation receives seeds and stage commitments. Neither is sent literal plaintext prompts, token IDs, or decoded output. Under protocol compliance, pseudorandom masks, mask non-retention, and non-collusion, the data plane removes the direct plaintext feed a provider could otherwise retain for training. This is a technical barrier rather than only a policy promise, but not a universal no-learning guarantee: metadata remains visible and collusion defeats the split.^[2]

Current implementation evidence

Nine warm Qwen2.5-0.5B runs used revision 277d19f on an Apple M5, with three repetitions per exact context after one warmup.^[3]

Input tokens	Output	TTFT	Client I/O
30	9	0.978 s	63.33 MB
63	16	1.374 s	126.29 MB
255	16	5.158 s	432.71 MB

All records ended with authoritative completion events, one retained model/body fingerprint pair, zero online Preparation protocol work, and zero plaintext prompt/token-byte audit counters. These are local CPU loopback measurements, not WAN, GPU, energy, price, or quality results.

Deployment gates

Preparation must be independently controlled or client-hosted; target-model quality must pass; the full cost ledger must beat explicit ceilings; and credential separation, erasure, restart behavior, and incident response must survive a pilot.

References

- [1] PLLM, Private inference.
- [2] PLLM, Security boundaries.
- [3] PLLM, Run your own benchmark; retained study data.